

SATYABRATA BEHERA

Software Developer | Data Engineer | ML Practitioner

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PROFESSIONAL SUMMARY

Results-driven Software Developer and aspiring ML Engineer with a strong foundation in Python, FastAPI, SQL, and Scikit-Learn. Demonstrated ability to engineer TDD-driven backend systems, design real-time data pipelines, and deploy CI/CD-automated workflows that measurably reduce error rates and improve system reliability.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, HTML5, CSS3

Frameworks: FastAPI, Flask, Scikit-Learn, Pandas, NumPy, Matplotlib, Streamlit, Pytest

Core Concepts: DSA, Machine Learning, Deep Learning, Generative AI, Backend Architecture, Speech & Audio Signal Processing, System Design

DevOps: Git, GitHub, CI/CD (GitHub Actions)

PROFESSIONAL EXPERIENCE

Generative AI Intern — Prodigy InfoTech (Remote)

May 2026 – Jun 2026

- Fine-tuned a **GPT-2** transformer model on a custom dataset to generate coherent, contextually relevant text mimicking a target style and structure, completing the internship with outstanding remarks.
- Built end-to-end text-generation pipelines covering data preprocessing, model fine-tuning, and prompt-based inference, strengthening applied experience with **transformer** architectures.

PROJECTS

Library Management Backend Engine

Jan 2026 – Present

Python • Pytest • GitHub Actions (CI/CD) • Custom File Storage

- Reduced manual processing errors by ~95% by engineering a modular **Python** backend automating inventory, borrowing, and fine calculations via a validated CSV storage system.
- Achieved ~90% test coverage with **TDD** via **Pytest**; cut integration failures by ~40% through **GitHub Actions** CI/CD automation.
- Eliminated database overhead by architecting a custom, schema-validated flat-file storage layer with full data integrity across concurrent operations.

Election Buddy

Dec 2025

Python • FastAPI / Flask • SQL • Git

- Architected a secure, role-based backend voting portal using **FastAPI** and **SQL** with cryptographic user authentication, achieving zero unauthorized-access incidents across all test scenarios.
- Engineered real-time vote-tallying pipelines with atomic SQL transactions, eliminating race conditions and ensuring 100% data integrity under concurrent load.
- Designed clean **REST** APIs with structured endpoint validation, reducing API error rates by ~35% versus initial prototype benchmarks.

AI-Powered Multi-Disease Diagnostic Tool

Nov 2025 – Dec 2025

Python • Scikit-Learn • Pandas • Streamlit • Gemini AI API

- Delivered ~85%+ average classification accuracy across 3 **Scikit-Learn** models (breast cancer, diabetes, heart disease) via end-to-end EDA, feature scaling, and hyperparameter tuning.
- Reduced diagnostic query time to under 5 seconds with an intuitive **Streamlit** UI abstracting complex ML inference for non-technical users.

EDUCATION

B.Tech — Electronics & Telecommunication Engineering

2024 – 2027

Veer Surendra Sai University of Technology (VSSUT), Odisha

ACHIEVEMENTS & CERTIFICATIONS

CS50x: Intro to Computer Science — Harvard University | Memory management, data structures, web architecture, C & Python.

Software Engineer Certification — HackerRank | Verified core problem-solving and Python proficiency.

Neo4j Certified Professional — Neo4j | Graph database design and query optimization.

Competitive Programming — LeetCode & Online Judges | Big-O optimization, advanced DSA, DP, graphs & greedy algorithms.

Full certification portfolio available on [LinkedIn](#) and [GitHub](#).